Suspended Sediment Particle Size Prediction

Random Forest, Support Vector Regression, and XGBoost

Problem statement

This project estimates two properties of suspended sediment particle-size distributions from environmental measurements: median particle diameter (d50) and distribution variance (sigma2). It compares three regression models using the workflow in the supplied Jupyter notebook.

Dataset and preparation

Dataset: the supplied *synthetic_data_improved.csv* contains 1,648 hourly rows, eight environmental predictors, and targets d50 and sigma2. Predictor values include missing entries. It is synthetic data, not verified field measurements or the original Stanford dataset.

Following the notebook, the time column is removed, incomplete rows are dropped (1,402 remain), and rows at or above the 99th percentile of either target are filtered out (1,384 remain). A shuffled 70/30 split with random_state=42 yields 968 training rows and 416 test rows. The four model inputs are S, ub, np, and T.

Approach and implementation

Separate models are fitted for d50 and sigma2. Random Forest uses 400 trees and reports out-of-bag scores. SVR uses a StandardScaler and a five-fold grid search over C and epsilon. XGBoost uses 300 trees, maximum depth 3, and learning rate 0.05. The notebook includes step-by-step preprocessing, feature-importance output, model metrics, a final comparison table, and charts. A Python script reproduces the workflow.

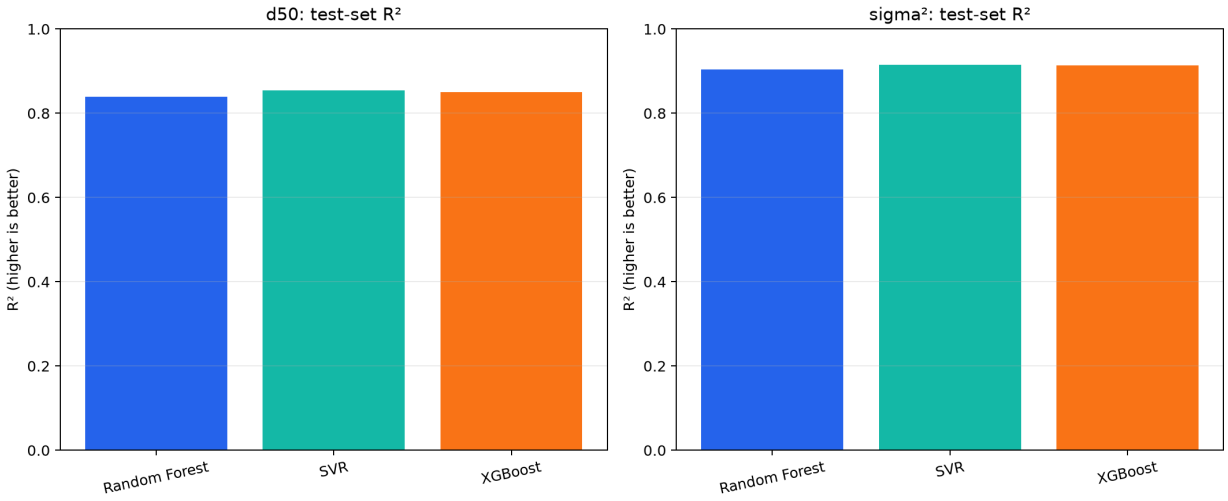
Test results

Target	Model	R^2	RMSE	MAE
d50	Random Forest	0.839727	15.769391	12.557029
d50	SVR	0.854116	15.044842	11.969561
d50	XGBoost	0.849638	15.274025	12.284159
sigma2	Random Forest	0.904587	7.931282	6.251181
sigma2	SVR	0.914782	7.495579	5.801557
sigma2	XGBoost	0.913948	7.532165	5.880333

Scores are computed on the notebook's held-out test set. Higher R^2 and lower RMSE/MAE indicate better fit on this split.

Findings and project notes

Results from the same workflow saved in the notebook



Conclusion

SVR has the highest test R^2 for both targets in this run: 0.854116 for d50 and 0.914782 for sigma2. Random Forest and XGBoost are also close on this synthetic dataset. These scores describe the notebook's selected test split only; they do not establish general performance on measured sediment samples.

Limitations

The dataset is synthetic, so its generated patterns may not represent real sediment behavior. The notebook computes the 99th-percentile cutoffs using the full dataset before splitting, which uses information from the eventual test targets and can make evaluation optimistic. Scores may vary slightly with library versions. A stronger future evaluation would define filtering rules using training data only and validate on independently collected observations.

Reproducibility

The repository includes the original step-by-step notebook as *notebooks/ML mini project.ipynb*, the CSV, requirements.txt, README, and run_project.py. From the repository root, install dependencies with `pip install -r requirements.txt` and run `python run_project.py`. To execute the notebook, open it from the project folder in Jupyter; it contains the result table and charts in its saved outputs.

Reference context

The project topic and model comparison relate to Galen Egan, "Machine learning for predicting suspended sediment particle size distributions," Stanford CS229 project report (2019). The dataset used here is the supplied synthetic file, and the results above are from this project's workflow, not Stanford's reported results. Stanford report.

Team members: _____ _____